How many solar panels do you need to charge a phone?

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CSS 290 – Software for Renewable Energy

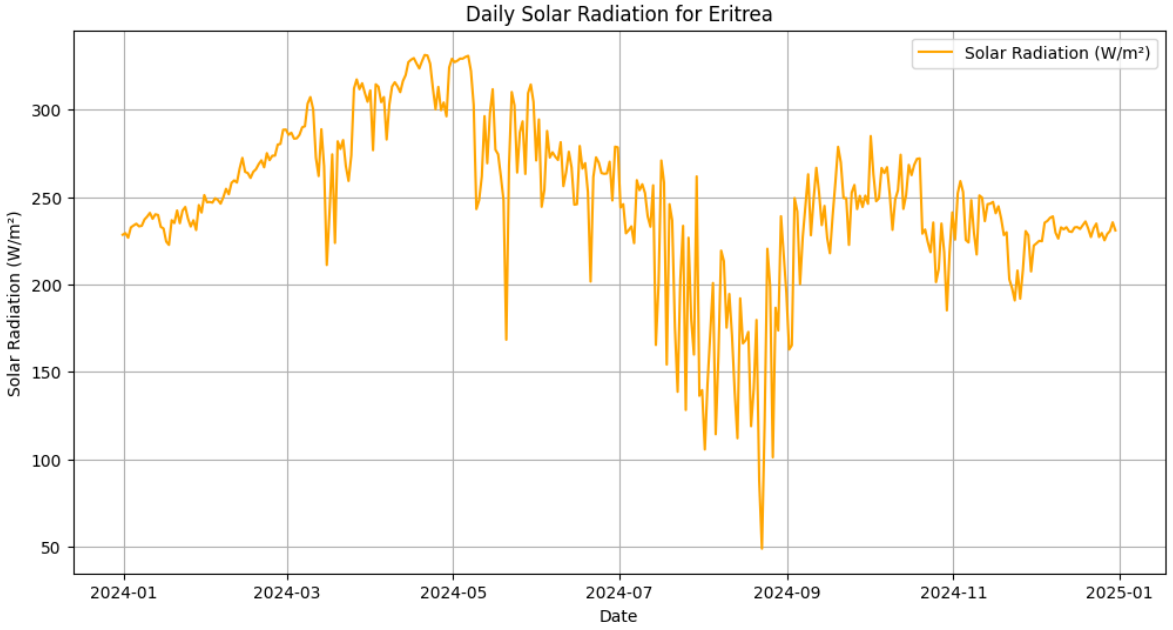
It depends on what you're charging

Phones that can store more charge will need more energy, which means more panels in our scenario.

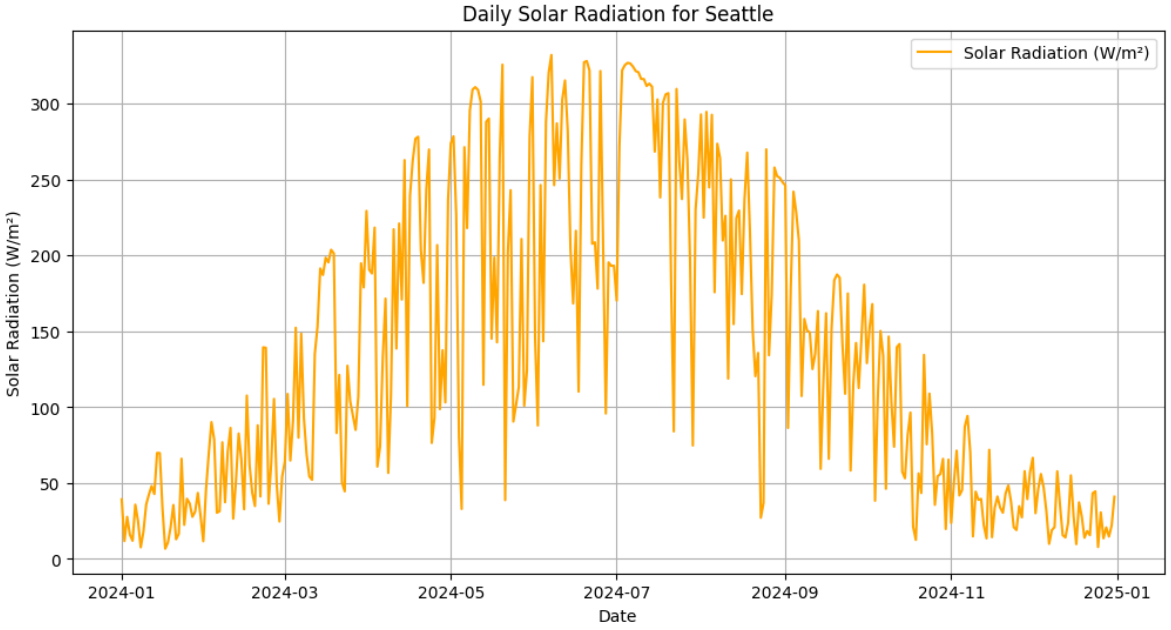
For example, the S25 Ultra can store over 500 mAh more than the iPhone 15 Pro Max, meaning it will take more solar energy to fully charge.

Model	Capacity
iPhone 15 Pro Max	4422 mAh
Nothing Phone (2)	4700 mAh
Huawei P60 Pro	4815 mAh
Xiaomi 14 Pro	4880 mAh
Samsung Galaxy S25 Ultra	5000 mAh

Eritrea



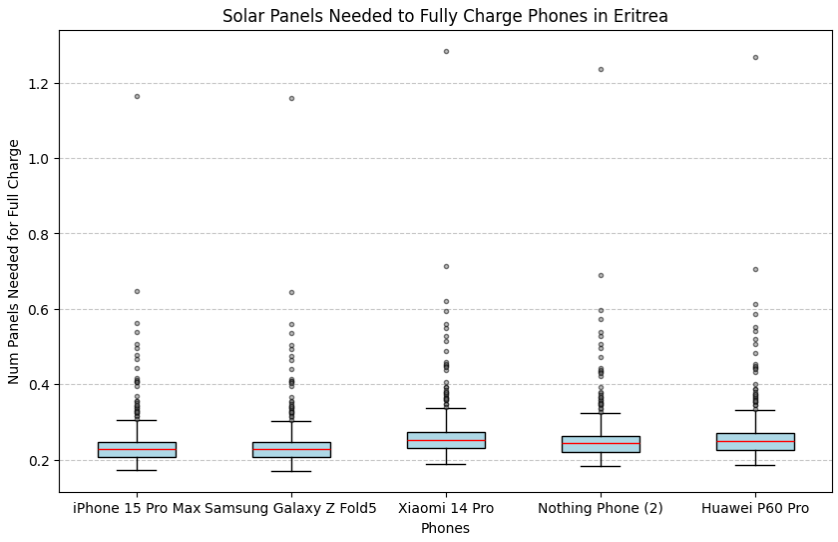
Seattle

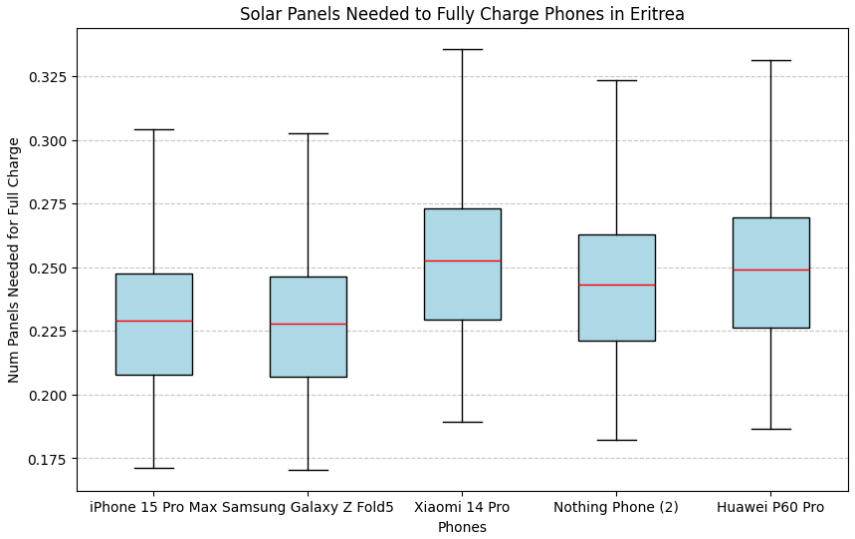


And it depends on where you live

Visualization

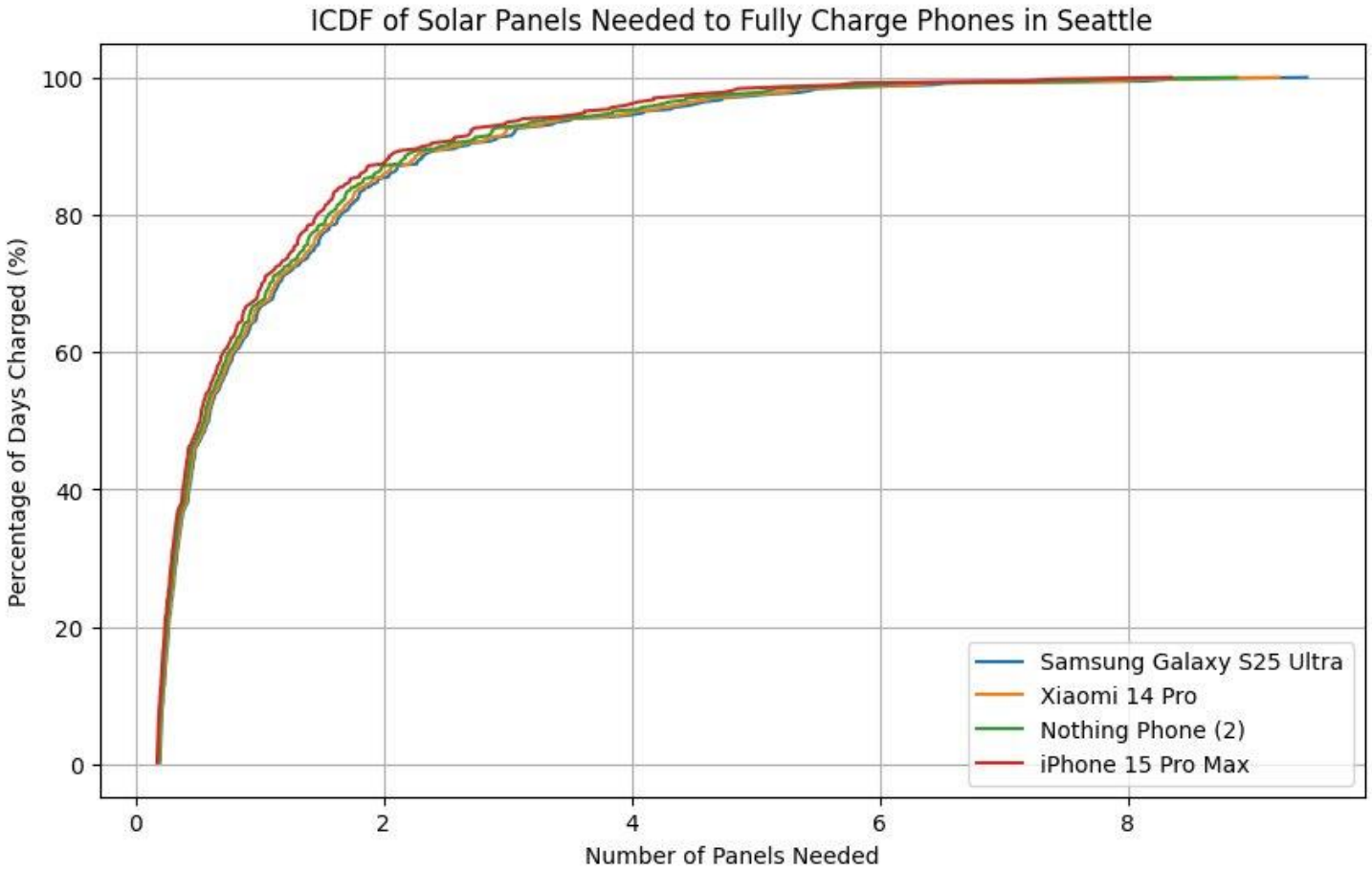
- Why choose a box and whisker graph?
  - Box and whisker graphs are a superior choice for data representation due to their ability to accurately convey key statistical measures.
  - Unlike bar graphs, which typically show only minimum and maximum values, box and whisker plots provide a comprehensive picture of data distribution by clearly displaying the median and range.
  - While initial data plotting may include individual points, these are often omitted in the final graph. This practice enhances clarity by preventing the dots from obscuring crucial statistical insights, thereby making the graph more effective and easier to interpret.





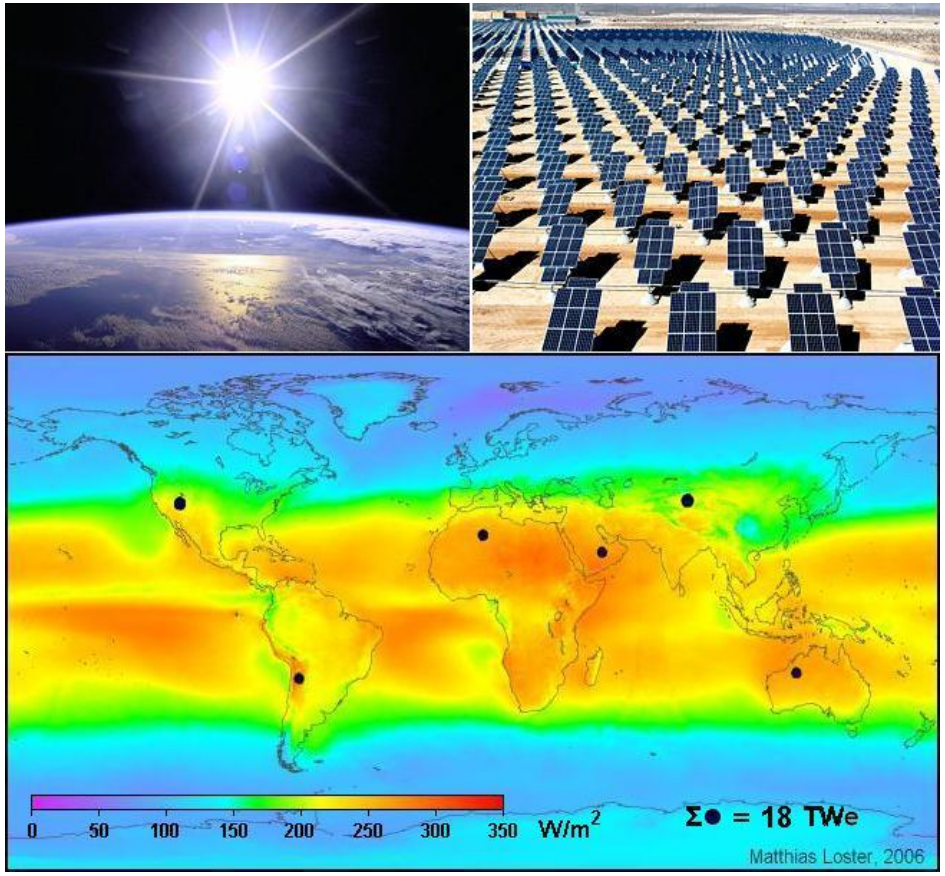
A different Visualization

If you look at this graph, you'll see that 90% of the year, you need only 2 panels to completely charge your phone.



Data Sources

- **Visual Crossings helped us grasp a Weather API for solar panel performance estimation**
  - *It delivers critical data: solar energy, temperature, and cloud cover.*

- **We've successfully used this API before.**
  - *Previously, it helped us predict solar farm output in Program 3.*

- **The API offers key advantages for our project.**
  - It's simple, accurate, and directly supports estimating solar charging needs by location year-round.